

Theoretical Analysis of the Effect of Ion Concentration in Solution Bulk and at Membrane Surface on the Mass Transfer at Overlimiting Currents

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Abstract—Overlimiting current modes are of considerable interest for the practice of electrodialysis (ED). However, the economical expedience of such ED modes is evident only for desalination of dilute solutions. Here, we show the theoretical analysis of the effect of concentration on the behavior of an ED cell with homogeneous ion-exchange membranes. The study is based on numerical solution of the two-dimensional system of coupled equations of Nernst–Planck–Poisson–Navier–Stokes. It is shown that as the electrolyte concentration in solution that enters the ED desalination chamber increases, the intensity of electroconvection decreases, which induces a decrease in the relative mass-transfer rate (the decrease in the ratio of current density to its limiting value). This effect is stronger in the region of high potential differences where the electroconvective instability of Rubinstein–Zaltzman is realized under the conditions of a nonuniform concentration field caused by solution desalination. In contrast, the increase in the counterion concentration at the membrane surface (associated with the increase in the surface charge) intensifies the electroconvection.

Keywords: electroconvection, overlimiting current mode, mathematical simulation, equations of Nernst–Planck–Poisson–Navier–Stokes

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INTRODUCTION

As was shown experimentally [1–4] and by mathematical simulations [5–7], electroconvection is the main mechanism of overlimiting transport in membrane systems in dilute electrolyte solutions. Electroconvection (EC) is the solution motion caused by the electric field action on the space charge localized near the ion-selective interface. The ion-selective surface is typical of ion-exchange membranes, electrodes, and microfluid devices with nanochannels [8]. To describe the EC mechanism, Dukhin and Mishchuk [9–10] introduced the concept of electroosmosis of the first and second kinds [9–11]. Electroosmosis of the first kind arises due to the action an external electric field on the equilibrium electric double layer (EDL) which exists even in the absence of electric current. Rubinstein and Zaltzman [5, 12] call this kind of osmosis (quasi)equilibrium. Electroosmosis of the second kind (or nonequilibrium electroosmosis [5, 13]) is associated with the electric field action on the extended space charge region (SCR) that appears at the overlimiting current densities. Electroosmosis of the second kind brings about sufficiently efficient solution mixing and intensifies the mass transport. For

membranes with certain optimization of properties, the mass transfer rate can several times exceed its value corresponding to the so-called limiting current density [2, 4, 6, 14].

The appearance of intense EC in membrane systems requires that three conditions be fulfilled [15–17]: (1) the presence of a sufficiently large space charge in solution; (2) the localization of charge in solution at a sufficient distance from the electrode/membrane surface; (3) the nonuniform distribution of the space charge to induce the tangential bulk force.

The fact that at $i \rightarrow i_{\text{lim}}$, the existence of the extended SCR that far exceeds the region of the equilibrium electric double layer was demonstrated for the first time by Rubinstein and Shtilman based on the numerical solution of equations of Nernst–Planck and Poisson (NPP) [18]. By considering a particle of ion-exchange resin with the curved surface, Dukhin and Mishchuk [9–11], proposed the following mechanism of EC development. The electric field should contain two components: its normal component creates the extended SCR due to the strong concentration polarization of solution near the interface, its tangen-

tial component sets in motion the charged liquid. For a membrane with the electrically nonuniform surface, the situation is analogous: the normal electric field induces the formation of the extended SCR, while the tangential field appears due to bending of current lines near the boundaries between the conducting and nonconducting surface areas (the funnel effect [19]).

In 2000, Rubinstein and Zaltzman [5] developed the model for describing the mass transport in solution between two uniform ion-exchange membranes in the absence of forced convection. They have found the numerical solution for equations of NPP and Navier–Stokes (NS) under the assumption of local electroneutrality in solution beyond the SCR by using a special condition of the electroosmotic slip at the boundary between the electroneutral region and the SCR. They have shown that the surface heterogeneity is not the necessary condition for the EC appearance. A characteristic feature of this system is its hydrodynamic instability at sufficiently high overpotentials. Several threshold potential drops were observed [5] which separated different phases in the development of electrokinetic instability. The first threshold potential drop corresponds to the transition from the stable state to unstable one. If the system is in the rest state (the flow velocity is equal to zero everywhere) and the potential drop is lower than its first threshold value, then the small perturbations of the fields of concentration or velocity die out because of viscosity and diffusion, and the system returns to its initial state. When the first threshold value of the potential drop is exceeded, the perturbations become sufficient for starting the process of generation of periodic paired vortices. After their appearance, such vortices are reorganized, their size and form change in the oscillating mode. However, after several oscillations, they are stabilized and stop to change in time. As the potential drop increases further, its second threshold value is reached and when the latter is exceeded, the oscillations of vortices are not dumped out but become chaotic.

The presence of all three modes of overlimiting transport in membranes was confirmed experimentally by Belova et al. [20].

The approaches like that of Rubinstein and Zaltzman [5] were used by Dydek et al. [21] and also by Green and Yossifon [22] for the description of electroconvective transport in microchannels. They used the slip condition on the internal boundary with SCR in the form of the formula of Helmholtz–Smoluchowski. The model based on the NNP and NS equations in which the counterion concentration was determined on the selectively permeable membrane surface, as in the 1D model of Rubinstein [18], and the flow of coions was equal to zero was considered by Demekhin et al. [23, 24], Pham et al. [25], and Druzgalski et al. [26].

Thus, in the membrane and micro-nanofluid systems without forced flow, two EC modes can be sin-

gled out. For curved or electrically nonuniform surfaces at underlimiting currents, the tangential electric field induces the steady electroosmotic transport described by Dukhin and Mishchuk [27–29]. Such mode is not realized on uniform ideally selective membranes in the absence of forced liquid flow: the EC appears as a result of hydrodynamic instability at overlimiting currents, as was shown by Rubinstein and Zaltzman [5]. For brevity sake, these two modes are sometimes referred to as the electrokinetic modes of Dukhin and Rubinstein, respectively [30]. Note that for nonideally selective membranes, the electrokinetic stability can arise also in the mode of (quasi)equilibrium electroosmosis, as was recently shown by Rubinstein and Zaltzman [12]. For an ideally selective membrane in the absence of forced flow, the electrochemical potential does not vary along the transversal coordinate in the membrane, as in the electric double layer. This has the stabilizing effect on the EC development.

The EC in ED channels with forced flow was studied by Urtenov et al. [7, 31] and Kwak et al. [6]. In such channels, the concentration is distributed nonuniformly along the channel length: as the solution moves between membranes, the electrolyte concentration decreases and the diffusion layer thickness increases. As a result, the current lines are concentrated closer to the channel entrance where system's resistance per unit of membrane surface is lower. In this case, the tangential bulk force arises and acts on the SCR at the depleted membrane surface even when the membrane is homogeneous. This force induces the stationary EC even at underlimiting current densities. This kind of EC can be considered as electroosmosis of the first kind [9, 10] or equilibrium EC [5, 12]. The difference from the mechanism described by Dukhin and Mishchuk lies in the fact that the tangential force required for electroosmosis of the first kind arises not because of the electrical nonuniformity of the surface but as a result of nonuniformity of the longitudinal concentration distribution.

In this mode of underlimiting currents, the bulk force is localized at a relatively small distance from the membrane where the viscous forces play the important role due to the adhesion condition. The contribution of EC to intensification of the mass transfer becomes considerable only at voltages corresponding to $i > i_{\text{lim}}$ at which the nonequilibrium EC develops (electroosmosis of the second kind). In this case, the SCR thickness sharply increases as compared with the equilibrium double layer to become of the order of several hundred nanometers. At such distances, the role of viscous forces decreases. Hence, the main contribution to the development of overlimiting transport is made by electroosmosis of the second kind.

In this study, we consider theoretically how the concentration at inlet to the electrodialyzer channel affects the intensity of EC development. It was found

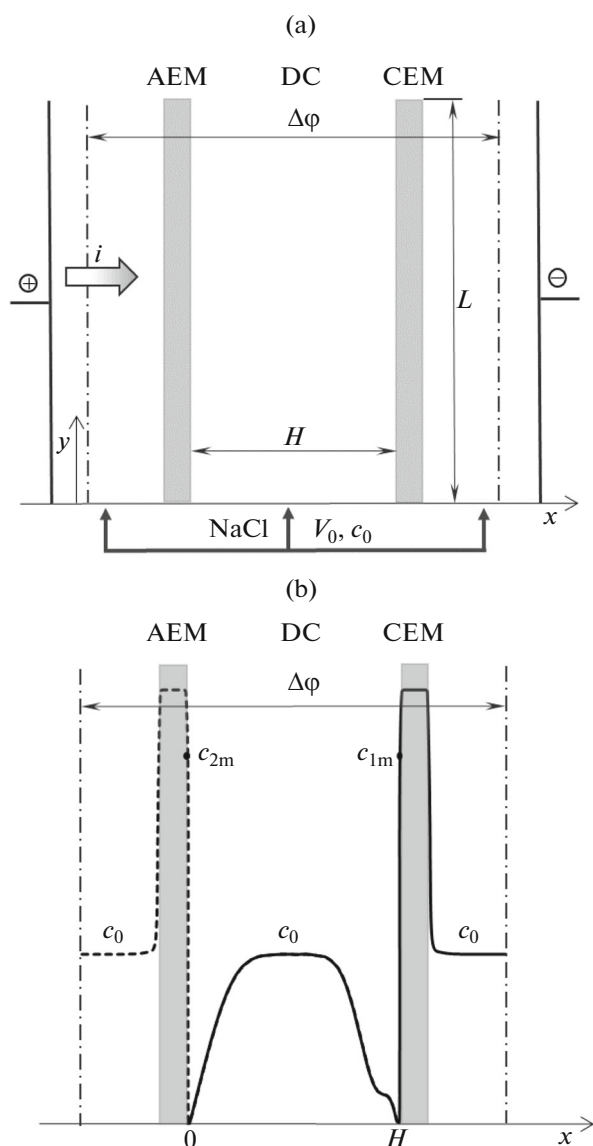


Fig. 1. Illustration of the modeled system; (a) two-dimensional geometry of the ED cell formed by anion-exchange (AEM) and cation-exchange (CEM) membranes; the desalination chamber (DC) and halves of concentration chambers (CC) are shown; polarizing electrodes are beyond the region under consideration; (b) concentration profiles in the central DC and in the adjacent concentration chambers. Solid line corresponds to cations, dashed line corresponds to anions; c_0 is the concentration in the flow core equal to the inlet concentration of electrolyte solution, c_{1m} and c_{2m} are the concentrations of cations at the CEM surface and anions at the AEM surface, preset by boundary conditions (7a) and (10a), respectively; $\Delta\phi$ is the potential difference between two equipotential planes (shown by dot-and-dash lines) in the neighboring concentration chambers. Concentration polarization in concentration chambers is ignored.

experimentally [2] that as the solution concentration increases, the contribution of EC to the overlimiting transport of ions decreases in favor of other mechanisms. However, as far as we know, the question on

the role of concentration was never considered in the literature.

The question on the role of solution concentration in the EC development is significant for the theory and practice of this phenomenon. Based on experimental data, the economical expedience of using intense (overlimiting) current modes in dilute solutions was proved [32]. However, it is important to know the trend of EC variations with the increase in solution concentration. This trend is ambiguous. On the one hand, as the concentration increases, the density of bulk electric charge at the membrane surface also increases, which should intensify the EC, on the other hand, the SCR approaches the interface, where the action of viscous forces that prevent the development of this phenomenon becomes more important.

MATHEMATICAL FORMULATION OF THE PROBLEM

The model used here includes the system of coupled equations NPP–NS, which for a binary electrolyte is written as follows [7, 33]:

$$\vec{j}_i = -\frac{F}{RT} z_i D_i c_i \nabla \phi - D_i \nabla c_i + c_i \vec{V}, \quad i = 1, 2, \quad (1)$$

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot \vec{j}_i, \quad i = 1, 2, \quad (2)$$

$$\varepsilon \varepsilon_0 \Delta \phi = -F(z_1 c_1 + z_2 c_2), \quad (3)$$

$$\vec{i} = F(z_1 \vec{j}_1 + z_2 \vec{j}_2), \quad (4)$$

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho_0} \nabla P + \nu \Delta \vec{V} \quad (5)$$

$$-\frac{F}{\rho} (z_1 c_1 + z_2 c_2) \nabla \phi, \quad \nabla \cdot \vec{V} = 0, \quad (6)$$

where D_i , z_i , c_i , and $\vec{j}_i$ are the diffusion coefficient, the charge number, the molar concentration, and the flow density of the i th ion, respectively; ϕ is the electric potential; t is the time; $\vec{V}$ is the solution flow rate; ρ_0 is the solution density (assumed to be constant); P is the pressure; ε_0 is the electric constant; ε is the relative dielectric permittivity of electrolyte solution (assumed to be constant); ν is the kinematic viscosity; F is the Faraday constant; R is the gas constant; T is the absolute temperature.

These equations are considered in two-dimensional geometry as applied to the desalination chamber of an ED cell containing an anion-exchange (AEM) and a cation-exchange (CEM) membranes, Fig. 1. Here, x is the coordinate transversal as regards the forced flow and varying from 0 (AEM interface) to H (CEM interface), Fig. 1; y is the longitudinal coordinate varying from 0 (channel inlet) to L (channel outlet). In the concentration chambers neighboring to the desalination chamber under consideration, the solution concentration is preset to be constant ($c = c_0$).

Here, P , $\vec{V}$, ϕ , $\vec{i}$, $\vec{j}_1$, $\vec{j}_2$, c_1 , and c_2 are unknown functions of t , x , and y . Equations of Nernst–Planck (1) together with equations of conservation (2), Poisson (3), and electric current (4) describe the fields of concentration and potential, whereas equations of Navier–Stokes (5) and continuity (6) describe the velocity field with regard to the effects of forced flow and electric bulk force. Two groups of equations are coupled by the flow rate in Eq. (1) and the concentration that determines the bulk force $\vec{f} = -F(z_1c_1 + z_2c_2)\nabla\phi$ in Eq. (5).

The electric mode is determined by setting the potential difference $\Delta\phi$ between two equipotential planes that are parallel to the membranes in concentration chambers to the left and to the right from the central desalination chamber. When calculating the voltammetric characteristic (VAC) of the electrodialysis channel, the $\Delta\phi$ scan rate was 0.01 V/s, which provided the quasistationary mode: decreasing the scan rate did not change the VAC and the concentration profiles.

At the conditional boundaries AEM/solution ($x = 0$, $y \in [0, L]$) and solution/CEM ($x = H$, $y \in [0, L]$), the concentration of counterions is set to be a constant value [18] N -times exceeding the concentration in solution bulk (N_a for AEM, N_c for CEM). On the same boundaries, the electric potential values are preset. The membranes are assumed to be impermeable for coions. The velocity of liquid at the membrane/solution boundaries is assumed to be zero (adhesion condition). Thus,

$$\left(\frac{\partial c_1}{\partial x} + \frac{F}{RT} z_1 c_1 \frac{\partial \phi}{\partial x}\right)(0, y, t) = 0, \quad (7), (7a)$$

$$c_2(0, y, t) = c_{2m} = N_a c_0,$$

$$V_x(0, y, t) = 0, \quad V_y(0, y, t) = 0, \quad (8)$$

$$\phi(0, y, t) = \frac{RT}{F} \ln N_a - \frac{1}{2} R_{ms} i_x(0, y, t), \quad (9)$$

$$\left(\frac{\partial c_2}{\partial x} + \frac{F}{RT} z_2 c_2 \frac{\partial \phi}{\partial x}\right)(H, y, t) = 0, \quad (10), (10a)$$

$$c_1(H, y, t) = c_{1m} = N_c c_0,$$

$$V_x(H, y, t) = 0, \quad V_y(H, y, t) = 0, \quad (11)$$

$$\phi(H, y, t) = \Delta\phi_{tot} - \frac{RT}{F} \ln N_c + \frac{1}{2} R_{ms} i_x(H, y, t), \quad (12)$$

$$\Delta\phi_{tot} = -\alpha t.$$

Here, α is the potential scan rate; c_0 is the concentration in the solution bulk for both desalination and concentration chambers. $\phi(0, y, t)$ is calculated as the sum of ohmic resistances of solutions in the left concentration chamber and in the AEM volume, to which the sum of two interfacial potential drops at AEM is added (Eq. (9)); $\frac{1}{2} R_{ms}$ is the ohmic resistance of this

part of the system. The interfacial potential drops are found from the Donnan relationship with disregard of the concentration gradient in the membrane. Analogously, we find Eq. (12) for $\phi(H, y, t)$, only now we start not from the left concentration chamber where the potential is zero, but from the right concentration chamber where the potential is $\Delta\phi$. By subtracting Eq. (12) from Eq. (9), we can obtain the following expression for the potential difference between two equipotential planes shown in Fig. 1:

$$\Delta\phi = -\frac{1}{2} R_{ms} i_x(0, y, t) - \frac{1}{2} R_{ms} i_x(H, y, t) - \frac{RT}{F} \ln N_a - \frac{RT}{F} \ln N_c + \phi(H, y, t) - \phi(0, y, t),$$

from which it is evident that $\Delta\phi$ represents the sum of (1) the ohmic potential drops in those parts of the system where the concentration polarization is absent and which have the overall resistance R_{ms} , (2) the Donnan potential drops at the interfaces of both membranes, (3) the potential drop in desalination chamber between points $x = 0$ and $x = H$.

The value R_{ms} determines the slope of the calculated voltammetric curve in its initial linear segment.

At the inlet to the channel ($x \in [0, H]$, $y = 0$), the electrolyte concentration is set constant c_0 ; the electric potential is determined from the condition of the zero tangential current density i_y through the inlet cross-section of the channel ($y = 0$) for any x ; the velocity profile is assumed to be parabolic

$$c_i(x, 0, t) = c_0, \quad i = 1, 2,$$

$$\frac{\partial \phi(x, 0, t)}{\partial y} = -\frac{RT}{F(z_1^2 D_1 + z_2^2 D_2)c_0} \times \left(z_1 D_1 \frac{\partial c_1(x, 0, t)}{\partial y} + z_2 D_2 \frac{\partial c_2(x, 0, t)}{\partial y} \right), \quad (13)$$

$$V_x(x, 0, t) = 0, \quad V_y(x, 0, t) = 6V_0 \frac{x}{H} \left(1 - \frac{x}{H}\right). \quad (14)$$

Here, V_0 is the average velocity of the forced solution flow.

At the outlet ($x \in [0, H]$, $y = L$), the sum of diffusion and migration components of the flows of cations ($i = 1$) and anions ($i = 2$) is assumed to be zero (which provides the zero tangential current density through the plane $y = L$); the velocity distribution is parabolic; the tangential derivatives of concentration and potential are assumed equal to zero:

$$\frac{\partial c_i}{\partial y}(x, L, t) = 0, \quad \frac{\partial \phi}{\partial y}(x, L, t) = 0, \quad (15)$$

$$\left(\frac{\partial c_i}{\partial y} + z_i c_i \frac{\partial \phi}{\partial y}\right)(x, L, t) = 0, \quad i = 1, 2, \quad (16)$$

$$V_x(x, L, t) = 0, \quad V_y(x, L, t) = 6V_0 \frac{x}{H} \left(1 - \frac{x}{H}\right). \quad (17)$$

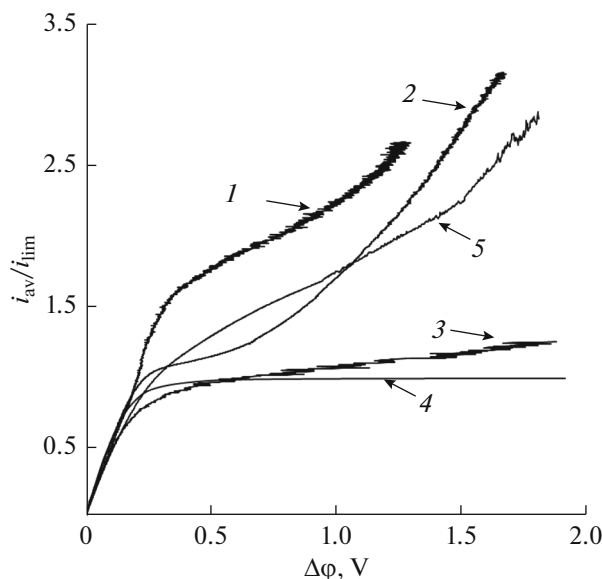


Fig. 2. (1–3) Experimental and (4, 5) calculated voltammetric characteristics of ED cells with ion-exchange membranes (1) Nafion, (2) CMX, (3) MK-40 are obtained for $c_0 = 20 \text{ mol/m}^3$, $V_0 = 3.6 \text{ mm/s}$, $H = 6.3 \text{ mm}$, $L = 20 \text{ mm}$, scan rate 0.01 mA/s ; (4) calculation in the framework of the convective-diffusion 2D model with NP-NS equations and the condition of local electroneutrality [36]; (5) calculation according to the basal model with equations NPP-NS (1)–(21). Adapted from [7].

For $t = 0$, the uniform distribution of concentration is preset (except for $x = 0$ and $x = H$, where the conditions (7a) and (10a) are fulfilled); the initial conditions agree with the other boundary conditions:

$$\begin{aligned} c_i(x, y, 0) &= c_0, \quad i = 1, 2, \quad 0 < x < H; \\ c_2(0, y, 0) &= N_a c_0, \quad c_1(H, y, 0) = N_c c_0, \end{aligned} \quad (18)$$

$$\begin{aligned} \varphi(x, y, 0) &= -\frac{1}{2} R_{ms} i_x(x, y, 0) \\ &+ (\Delta\varphi + R_{ms} i_x(x, y, 0)) \frac{x}{H}, \quad 0 < x < H, \end{aligned} \quad (19)$$

$$\begin{aligned} \varphi(0, y, 0) &= \frac{RT}{F} \ln N_a - \frac{1}{2} R_{ms} i_x(0, y, 0), \\ \varphi(H, y, 0) &= \Delta\varphi - \frac{RT}{F} \ln N_c + \frac{1}{2} R_{ms} i_x(H, y, 0), \\ V_x(x, y, 0) &= 0, \quad V_y(x, y, 0) = 6V_0 \frac{x}{H} \left(1 - \frac{x}{H}\right). \end{aligned} \quad (20)$$

The average current density in the channel is calculated according to formula

$$i_{av} = \frac{1}{HL} \int_0^L \int_0^H i_x dx dy. \quad (21)$$

The formulated problem was solved by the finite element method by using Comsol Multiphysics 5.1 software package. The solution algorithm is described briefly in [34].

Figure 2 shows the experimental and calculated voltammetric characteristics of individual ion-exchange membranes; the current density i_{av} averaged according to Eq. (21) was normalized to the limiting current density i_{lim} calculated by the Leveque equation [35]

$$i_{lim} = \frac{FDc_0}{H(T_1 - t_1)} \left[1.47 \left(\frac{H^2 V_0}{LD} \right)^{1/3} - 0.2 \right]. \quad (22)$$

Equation (22) is the asymptotic solution of the two-dimensional problem of convective diffusion in a channel between two plane parallel plates (in our case, membranes) with the established laminar solution flow. This equation allows i_{lim} to be exactly calculated if the channel length is sufficiently small ($L < 0.02 V_0 H^2 / D$).

As follows from Fig. 2, the value i_{lim} found based on the Leveque equation (22) corresponds exactly to the calculated current density at which the saturation of diffusion transport takes place and the VAC reaches a horizontal plateau if the equations of Nernst–Planck and Navier–Stokes and the conditions of local electroneutrality are used [36]. At the same time, the VAC found from the solution of coupled equations of Nernst–Planck–Poisson and Navier–Stokes (1)–(21) demonstrates no current limit. Like the experimental curves, the calculated VAC (curve 5 in Fig. 2) rises much higher than i_{lim} , the reason for this is the electroconvective transfer of ions. The initial linear segment of VAC is followed by an inclined plateau at $\Delta\varphi$ values corresponding to the attainment of i_{lim} . Initially, the current in the plateau increases monotonously and the curve is smooth. This part of curve corresponds to the stable EC developing by the mechanism analogous to that of Dukhin–Mishchuk [9, 29]: The appearance of the tangential component of the bulk electric force is associated with variation of the local current density along the transversal coordinate. However, in contrast to the “classical” treatment, the current density along the membrane surface varies not as a result of electrical nonuniformity of the membrane surface (in the model, the membrane is assumed to be homogeneous) but as a result of nonuniformity of the concentration field [7]: the current lines are concentrated in the vicinity of the channel inlet because, as solution moves, its desalination takes place and the specific resistance increases. This VAC segment is followed by the segment in which oscillations of current/potential appear. The appearance of oscillations is associated with the transition to the mode of electroconvective instability of Rubinstein–Zaltzman [5]. The sharper oscillations and the steeper ascend of the VAC are associated with the beginning of intense EC in the vicinity of AEM where this phenomenon arises at the higher current density (because the limiting current density through the AEM in NaCl solution is approximately 1.5-times higher than through the CEM).

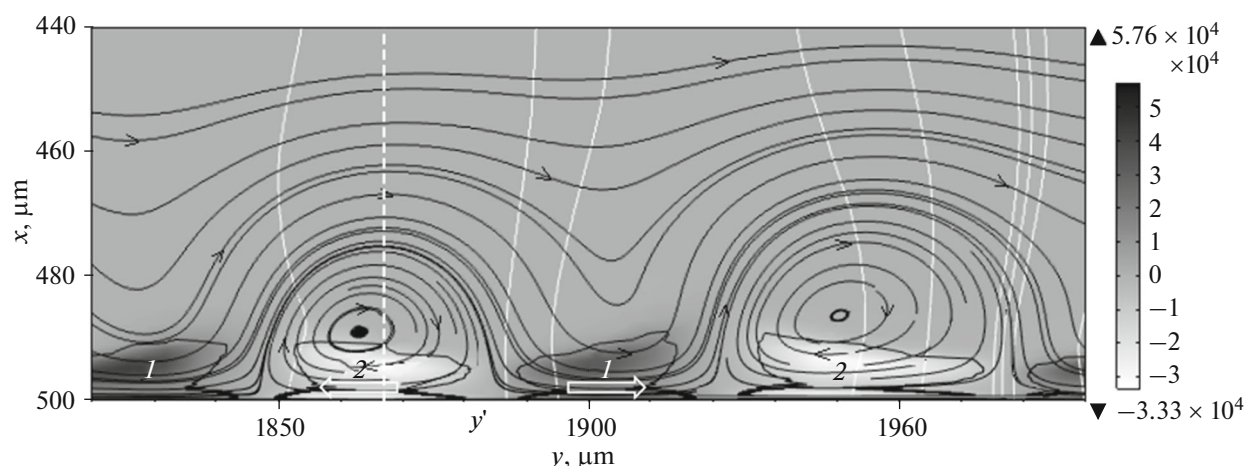


Fig. 3. Distribution of the tangential component of electric force F_{ey} (the magnitude is shown by shades of gray; the direction is shown by arrows), solution flow lines (black lines), electric field lines (white lines), cross-section y' (white dashed line) in the overlimiting state. Calculation for $\Delta\phi = 0.8$ V, $c_0 = 0.01$ mol/m³. (1) Regions where $F_{ey} > 10^4$ N/m³, (2) regions where $F_{ey} < -10^4$ N/m³.

It is also seen that VACs measured for various membranes differ considerably from one another. The reason is the different properties of the membrane surface: the different degree of its geometrical and electrical nonuniformity, the different degree of hydrophobicity. The “basal” model (Eqs. (1)–(21)) developed for a certain idealized membrane adequately describes on the qualitative level the real behavior of IEM but ignores the concrete features of each membrane.

THE EFFECT OF THE INLET ELECTROLYTE CONCENTRATION

The calculations were carried out for the following values of parameters: the intermembrane distance (channel width) $H = 0.5$ mm (typical of real ED channels) channel length $L = 2$ mm, average velocity of solution flow $V_0 = 0.8$ mm/s, temperature $T = 298$ K, electrolyte solution density $\rho_0 = 1002$ kg/m³, coefficient of dynamic viscosity $\eta = 0.8937 \times 10^{-3}$ Pa s, the diffusion coefficients of cations and anions $D_1 = 1.33 \times 10^{-9}$ m²/s and $D_2 = 2.05 \times 10^{-9}$ m²/s, respectively, the ratio of counterion concentrations at the boundary with AEM or CEM and at the channel inlet $N_a = c_2(0, y, t)/c_0 = c_{2m}/c_0 = 1$, $N_c = c_1(H, y, t)/c_0 = c_{1m}/c_0 = 1$ (Fig. 1).

The small channel length (2 mm) and the low values of the inlet concentration c_0 (from 0.01 to 10 mol/m³) are associated with the necessity of using a sufficiently detailed calculation network; within the EDL region (thickness L_D) the network should have no less than 20 points in the direction normal to the membrane surface [34, 37]. As the concentration increases, the EDL thickness at the membrane surface decreases in inverse proportion to $\sqrt{c_0}$; for $c_0 = 10$ mol/m³, $L_D \approx 3$ nm. The calculations with the values of concentration and channel length used in the experiment require

the unfeasible increase in the computation complexity. Karatay et al. (Stanford University, USA) [37] assessed the computation complexity for a similar problem and showed that it rapidly increases with the decrease in the dimensionless Debye length $\varepsilon = L_D/H$. For $c_0 = 0.01$ mol/m³ and $H = 0.5$ mm, as in the case of the minimum concentration used in our study, $L_D \approx 100$ nm and $\varepsilon = 2 \times 10^{-4}$. For $c_0 = 10$ mol/m³, $\varepsilon = 6 \times 10^{-6}$. Note that calculations in [37] were carried out for $\varepsilon = 10^{-3}$. On the other hand, the attempts to carry out an accurate comparison of calculation and experiment are confronted by the fact that as we decrease the inlet concentration, the errors of measurements considerably increase: the dissociation of water at the membrane/solution interfaces goes out of control, which leads to considerable involvement of ions H^+ and OH^- in the charge transfer. In addition, it is difficult to decrease the geometrical parameters of the experimental cell. Thus, the parameters in our study are the compromise determined by limitations of both experiment and calculations.

The main results are shown for the potential difference $\Delta\phi = 0.8$ V, where EC proceeds in nonequilibrium mode by the mechanism of electroosmosis of the second kind.

In the overlimiting state at the electrolyte-solution/membrane interface, the electroconvective vortices develop; moreover, the tangential component of the electric force F_{ey} within the closed vortices (rotating anti-clockwise) is negative being directed to the channel inlet (Fig. 3). It is positive in those regions where the forced flow destroys the vortices that rotate anti-clockwise. The vortex structure pattern changes depending on the applied potential difference and the time. For comparing the characteristics, including the forces at different c_0 , we chose the sections $y = y'$ near the outlet from the channel in which the vortices have the similar structure. Namely,

Table 1. Results of calculations of the tangential bulk force per volume unit F_{ey} and per surface unit $\hat{F}_e$, and also of the SCR thickness l_{SCR} for the potential difference $\Delta\phi = 0.8$ V in the cross-section y' for different inlet concentrations of electrolyte c_0

c_0 , mol/m ³	y' , μm	F_{ey} , N/m ³	l_{SCR} , μm	$\hat{F}_e = F_{ey}l_{SCR}$, N/m ²	$\hat{F}_e(c_0)/\hat{F}_e(0.1c_0)$
10	1926.5	-2.802×10^7	0.41	-11.49	3.57
1	1421.9	-3.097×10^6	1.04	-3.22	5.11
0.1	1644.9	-2.566×10^5	2.47	-0.63	3.32
0.01	1868.0	-3.533×10^4	5.48	-0.19	

in these sections, the force F_{ey} as the function of x and y has a local minimum (Fig. 3). Table 1 shows the thus obtained characteristics.

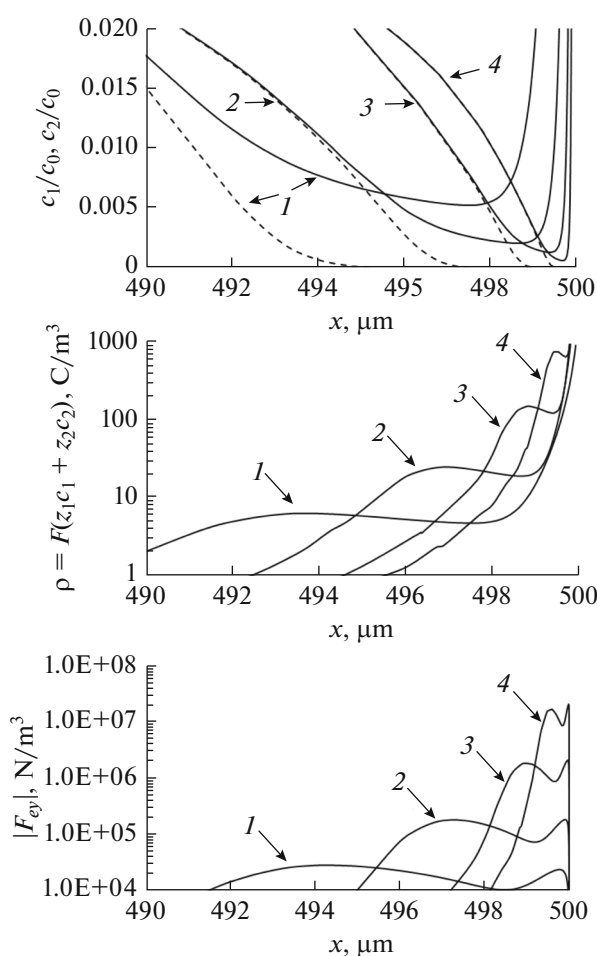


Fig. 4. (a) Dimensionless concentration profiles (solid lines correspond to cation concentration, dashed lines correspond to anion concentration); (b) space charge ρ as a function of the distance to the membrane; (c) absolute magnitude of the tangential component of the bulk force $|F_{ey}|$ in cross-section y' at the CEM. Calculations are carried out for the overlimiting state $\Delta\phi = 0.8$ V and different concentrations c_0 : (1) 0.01 mol/m³; (2) 0.1 mol/m³; (3) 1 mol/m³; (4) 10 mol/m³.

Figure 4 shows the concentration profiles, the charge density, and the absolute magnitude of the tangential component of electric force F_{ey} in the overlimiting state (for $\Delta\phi = 0.8$ V) for different values of the inlet concentration c_0 in the y' cross-section at the CEM. Figures 4a and 4b show that as the concentration c_0 increases, the SCR that determines the electric force narrows down. As a result, the distance from the local maximum of force $|F_{ey}|$ to the membrane surface decreases (Table 1). Moreover, as the concentration increases, the force also increases in the absolute value $|F_{ey}|$ (Fig. 4c).

The density of the bulk electric force is proportional to the product of the space charge density ρ and the electric field strength E : $F_e = \rho E$. Insofar as within the SCR the coion concentration is negligibly small [18, 38], $\rho = c_1 F$, then, $F_e = c_1 F E$. In the extended space charge region, the diffusion contribution to the transport of counterions is very small, the concentration of counterions is approximately constant and equal to c_{1s} ; hence, the equation of Nernst–Planck is simplified: $i = D_1 c_{1s} (F^2/RT) E$. The current density is close to the limiting value; hence, by combining the latter equation with the equation of Peers $i_{lim} = \frac{Dc_0 F}{\delta_N (T_1 - t_1)}$, we can estimate $F_e \approx 2c_0 RT/\delta$ [38, 39].

The bulk force per membrane surface unit can be assessed as the product $\hat{F}_e = F_e l_{SCR}$, where l_{SCR} is the thickness of the extended SCR. According to [40], l_{SCR} has the order of magnitude $l_{SCR} \sim \delta (L_D/\delta)^{2/3}$, where δ is the diffusion layer thickness and $L_D = \sqrt{\frac{RT\epsilon\epsilon_0}{2c_0 F^2}}$ is the

Debye length. Then, $\hat{F}_e \sim (c_0/\delta)^{2/3}$. According to this estimate, as the concentration increases 10-fold, the force applied to the extended SCR calculated per membrane surface unit increases approximately 4.6-fold.

Table 1 shows the results of numerical calculation of the bulk force and the ratio of these forces $\hat{F}_e(c_0)/\hat{F}_e(0.1c_0)$ as the concentration increases

10-fold. The thickness l_{SCR} was determined by the point of charge density maximum. It is seen that the ratio of forces (the minimum of $\hat{F}_e(c_0)/\hat{F}_e(0.1c_0)$ is 3.32, its maximum is 5.11) corresponds to the aforementioned estimates. The scatter of the force ratio is explained by the fact that system's state at a fixed potential difference is unstable and the characteristics have random values which however fit a certain finite interval.

Everything aforesaid about the random character of the results of calculations is well illustrated by voltammetric curves shown in Fig. 5 for different concentration values c_0 . The current density is normalized to the limiting current density proportional to the concentration, according to the Levac equation (Eq. (22)).

From this figure, it follows that before the threshold potential drop ($\Delta\phi_{\text{cr}}$ is about 0.7 V), the curves are smooth and the current increases monotonously with the increase in $\Delta\phi$. In this region, EC is stable and develops by the mechanism of Dukhin–Mishchuk (with correction for the fact that the quasi-equilibrium character of EC is caused by nonuniformity of the concentration field, as shown above, rather than by the surface heterogeneity). For $\Delta\phi > \Delta\phi_{\text{cr}}$, the EC becomes unstable: when the potential difference is scanned slowly (10 mV/s), the current density changes nonmonotonously, demonstrating oscillations with the amplitude increasing with the increase in $\Delta\phi$. The brief analysis of these oscillations and the comparison of their parameters with experimental values were carried out in [7]. It was shown that there is qualitative agreement but the parameters of oscillations depend substantially on the membrane surface properties (electrical and geometrical microheterogeneity of the surface, its hydrophilic-hydrophobic balance). The basal model [7] used in this study describes a certain “ideal” membrane with 100% selectivity and the homogeneous smooth surface. The theoretical analysis of the effect of membrane properties on its behavior at overlimiting currents is the subject of our subsequent studies.

As to the effect of concentration c_0 on the VAC, it appears even in the underlimiting state ($i/i_{\text{lim}} < 1$) and strengthens with the increase in the current density. As the concentration c_0 increases, the dimensionless current density decreases (at fixed $\Delta\phi$), and the threshold value of the potential difference at overlimiting transfer ($\Delta\phi_{\text{cr}}$) increases.

Table 2 shows the time-averaged dimensionless current density at the potential $\Delta\phi = 0.9$ V calculated for different values of inlet concentration. Such a potential difference corresponds to the underlimiting current but the latter is higher than $\Delta\phi_{\text{cr}}$. Table 2 shows that the difference between the current densities is sufficiently large.

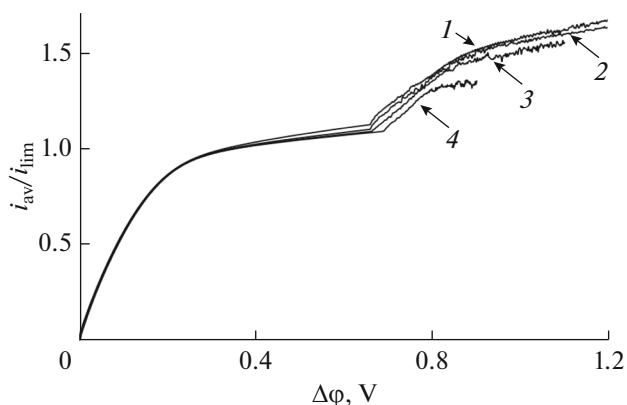


Fig. 5. VAC calculated for different inlet concentrations c_0 : (1) 0.01 mol/m³; (2) 0.1 mol/m³; (3) 1 mol/m³; (4) 10 mol/m³.

Figure 6 shows the variations in the structure of the electroconvective vortex region with the increase in the concentration c_0 .

THE EFFECT OF COUNTERION CONCENTRATION AT THE MEMBRANE SURFACE

For studying how counterion concentration determined by the boundary conditions (7a) and (9a) at the membrane surface affects the membrane behavior, we carried out the calculations for the fixed $c_0 = 1$ mol/m³. For simplicity sake, the parameters N_a and N_c that determine the ratio of counterion concentration at the boundary with AEM ($x = 0$) or CEM ($x = H$) to its value at the channel inlet were assumed to be equal to

Table 2. Time-averaged dimensionless current density at $\Delta\phi = 0.9$ V calculated for different values of the inlet concentration

Inlet concentration c_0 , mol/m ³	Time-averaged dimensionless current density at $\Delta\phi = 0.9$ V, $\bar{i}_{\text{av}}/i_{\text{lim}} = \left(\frac{1}{\Delta t} \int_{t_0}^{t_0+\Delta t} i_{\text{av}}(t) dt \right) / i_{\text{lim}}$ $t_0 = 90$ s, $\Delta t = 3$ s
0.01	1.533
0.10	1.518
1.00	1.482
10.00	1.343

i_{av} is the membrane-length averaged current density in moment t , Eq. (21).

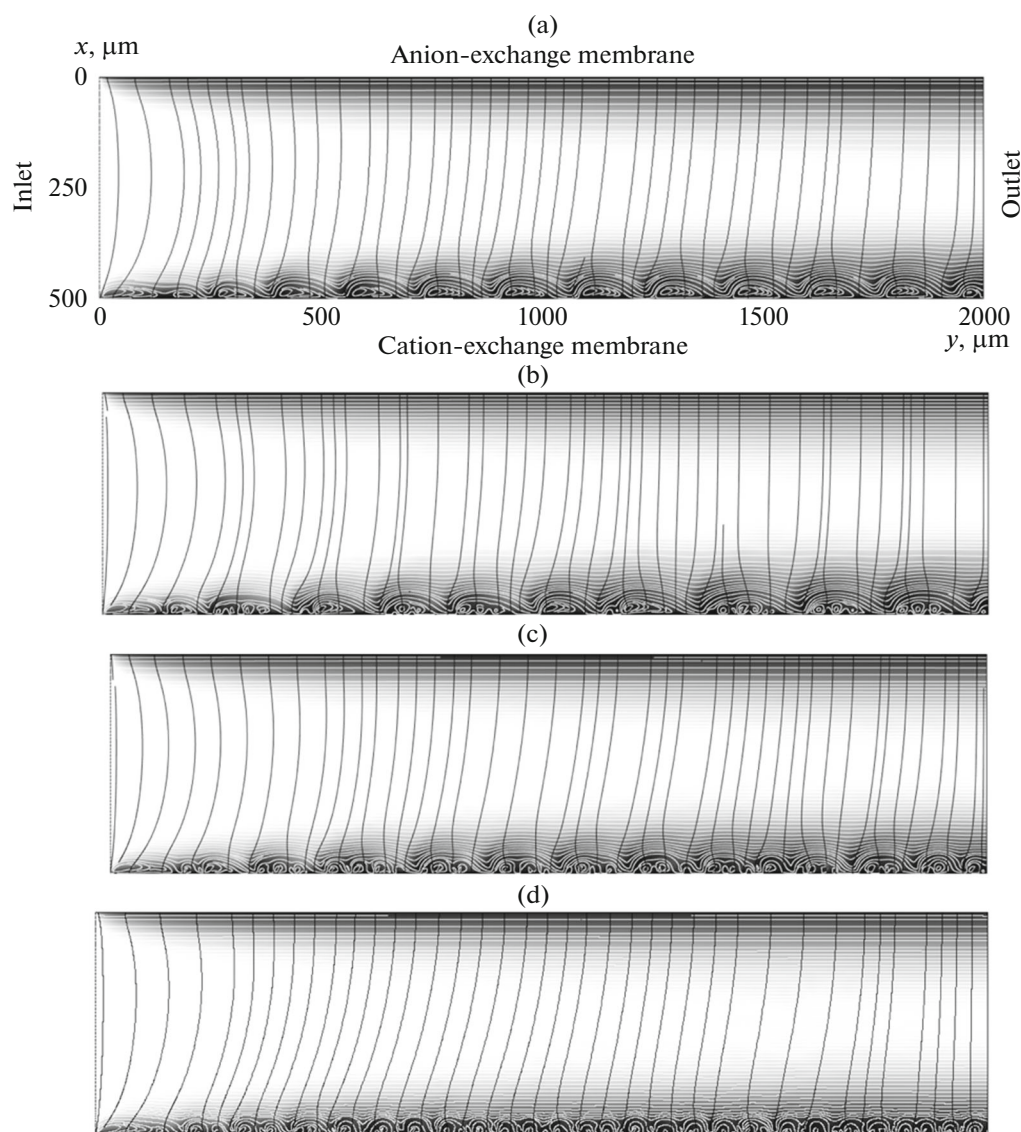


Fig. 6. Distribution of cation concentration (shades of gray), solution flow lines (white lines), electric current lines (black lines) for $\Delta\phi = 0.9$ V and different concentrations c_0 : (a) 0.01 mol/m^3 ; (b) 0.1 mol/m^3 ; (c) 1 mol/m^3 ; (d) 10 mol/m^3 .

one another and take the values 10, 1, and 0.1. The other parameters remained unchanged.

Figure 7 shows the calculated VAC. It is seen that as $N_a = N_c$ increases, the tendency for the increase in the current density appears at the fixed $\Delta\phi$. However, at $N_a = N_c = 10$ and $N_a = N_c = 1$, the VAC curves are indistinguishable. At $N_a = N_c = 0.1$, the curve is localized somewhat lower; moreover, the threshold potential difference $\Delta\phi_{cr}$ is somewhat higher than for $N_a = N_c = 1$.

In Fig. 7, besides curves 1, 2, and 3 calculated with the use of the condition of the Dirichlet type according to which the concentration at the solution/membrane interface is fixed, we added curve 4 calculated with the boundary condition of the Neumann type [7]

where the normal derivative of counterion concentration is zeroed out:

$$\frac{\partial c_1}{\partial x}(H, y, t) = 0, \quad \frac{\partial c_2}{\partial x}(0, y, t) = 0. \quad (23)$$

Physically this means that the boundary of the solution region under consideration is transferred to the plane in which the counterion concentration reaches its minimum (Fig. 8a). This minimum is located at a distance of the Debye length from the membrane surface, determined from the counterion concentration in the extended SCR [38]. It is evident that curve 4 is considerably lower than the other curves that provide more adequate information on the behavior of the system. In this case, the ratio of N_c depends on the potential shift and the coordinate y . For $\Delta\phi =$

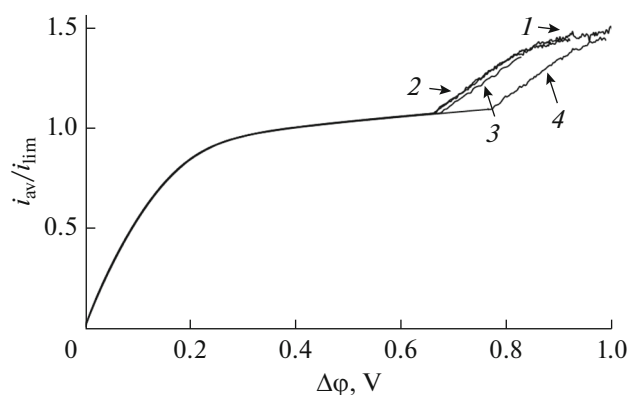


Fig. 7. VAC calculated for $c_0 = 1 \text{ mol/m}^3$ and different values of $N_a = N_c$: (1) 10; (2) 1; (3) 0.1; (4) condition (23).

0.8 V, N_c varies from 0.0003 to 0.0018 along the membrane surface, except for a small region at the channel inlet. Hence, the region of quasi-equilibrium double layer at the membrane is eliminated from consideration and the adhesion condition is transferred to the external boundary of this region. It is evident that such simplification considerably decreases the calculated intensity of the EC. On the other hand, the numerical solution can be obtained on a network with the smaller number of elements and the lower calculation complexity.

Figure 8 shows the dimensionless concentration profiles and the space charge density in the cross section $y = 1800 \text{ }\mu\text{m}$ for the potential difference $\Delta\phi = 0.6 \text{ V}$. For $N_a = N_c = 10$ and $N_a = N_c = 1$, the space charge

curves approximately coincide; for $N_a = N_c = 0.1$, the curve maximum is located closer to the membrane; for condition (24), it is much closer to the membrane.

In [38], based on the solution of the boundary condition for NPP equations in the one-dimensional case, it was shown that the local maximum of the space charge density with the high degree of accuracy is independent of the interfacial concentration of counterions. Calculations carried out for $N_a = N_c = 10$; 1; 0.1 and for condition (23) showed that the values of local minimums differ by no more than 1%. Nonetheless, the threshold voltage of the transition to the mode of Rubinstein–Zaltzman instability depends on N_a and N_c , because the latter affect the space-charge localization: for $N_a = N_c = 10$ and 1 the minimums approximately coincide; for $N_a = N_c = 0.1$, a small shift towards the membrane is observed; for condition (23), the shift becomes considerable.

CONCLUSIONS

As the electrolyte concentration in solution (c_0) supplied to the flow-through ED cell increases, the SCR size in depleted solution at the membrane surface decreases; however, the charge density and the tangential component of the bulk electric force increase. These two factors exert the opposite effects on the development of EC in depleted solution at the membrane surface. The results of numerical calculations showed that the factor responsible for the decrease in the SCR size dominates: as the concentration increases, the intensity of overlimiting transport determined by the ratio i/i_{lim} decreases at the fixed potential

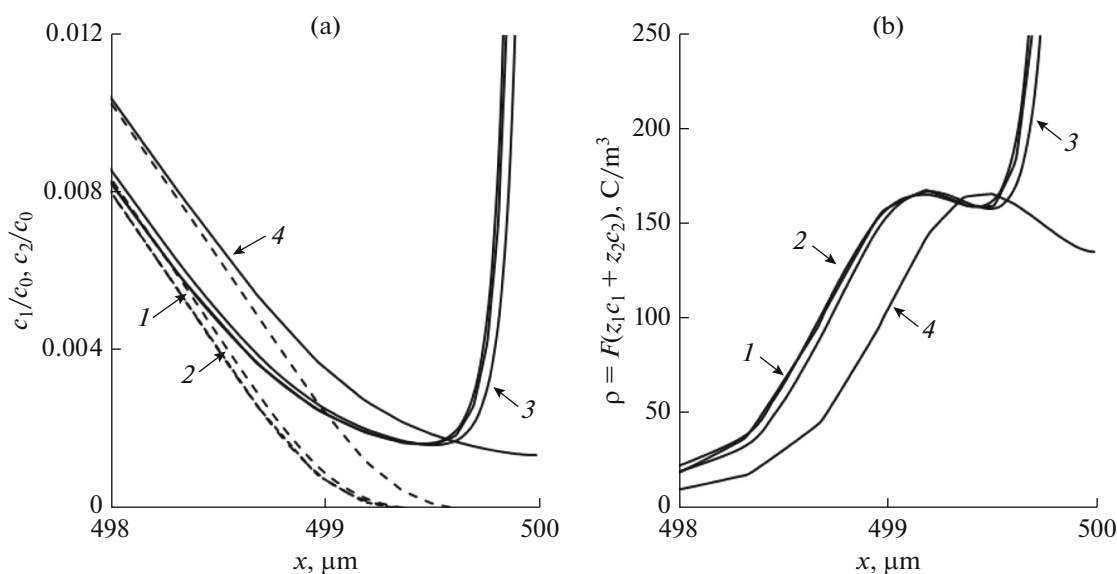


Fig. 8. (a) Dimensionless concentration profiles near the CEM surface (solid lines are cation concentrations; dashed lines are anion concentrations) in the cross-section $y = 1800 \text{ }\mu\text{m}$, (b) space charge density in the same cross-section. Calculations for $c_0 = 1 \text{ mol/m}^3$, potential difference $\Delta\phi = 0.6 \text{ V}$, and different $N_a = N_c$: (1) 10; (2) 1; (3) 0.1; (4) condition (23).

difference. However, the stable EC in the Dukhin–Mishchuk mode depends weakly on c_0 ; the effect of concentration growth strengthens with the increase in the potential difference. The higher c_0 , the larger the threshold potential difference ($\Delta\phi_{cr}$) at which the transition from the Dukhin–Mishchuk mode to the mode of Rubinstein–Zaltzman is observed. In the instability mode (oscillations appear in the voltammogram), the ratio i/i_{lim} considerably decreases with the increase in c_0 . The increase in counterion concentration at the membrane interface (c_m) has the opposite effect on EC: as c_m increases, $\Delta\phi_{cr}$ decreases and the EC intensity increases.

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REFERENCES

- Maletzki, F., Rösler, H.W., and Staude, E., Ion transfer across electrodialysis membranes in the overlimiting current range: stationary voltage characteristics and current noise power spectra under different conditions of free convection, *J. Membr. Sci.*, 1992, vol. 71, p. 105.
- Pismenskaya, N.D., Nikonenko, V.V., Belova, E.I., Lopatkova, G.Yu., Sistat, P., Pourcelly, G., and Larshe, K., Coupled convection of solution near the surface of ion-exchange membranes in intensive current regimes, *Russ. J. Electrochem.*, 2007, vol. 43, p. 307.
- Rubinstein, S.M., Manukyan, G., Staicu, A., Rubinstein, I., Zaltzman, B., Lammertink, R.G.H., Mugele, F., and Wessling, M., Direct observation of a nonequilibrium electro-osmotic instability, *Phys. Rev. Lett.*, 2008, vol. 101, 236101.
- Vasil'eva, V.I., Zhiltsova, A.V., Malykhin, M.D., Zabolotskii, V.I., Lebedev, K.A., Chermit, R.Kh., and Sharafan, M.V., Effect of the chemical nature of the ionogenic groups of ion-exchange membranes on the size of the electroconvective instability region in high-current modes, *Russ. J. Electrochem.*, 2014, vol. 50, p. 120.
- Rubinstein, I. and Zaltzman, B., Electro-osmotically induced convection at a permselective membrane, *Phys. Rev. E*, 2000, vol. 62, no. 2, p. 2238.
- Kwak, R., Pham, V.S., Lim, K.M., and Han, J., Shear flow of an electrically charged fluid by ion concentration polarization: scaling laws for convection vortices, *Phys. Rev. Lett.*, 2013, vol. 110, 114501.
- Urtenov, M.K., Uzdénova, A.M., Kovalenko, A.V., Nikonenko, V.V., Pismenskaya, N.D., Vasil'eva, V.I., Sistat, P., and Pourcelly, G., Basic mathematical model of overlimiting transfer enhanced by electroconvection in flow-through electrodialysis membrane cells, *J. Membr. Sci.*, 2013, vol. 447, p. 190.
- Daiguji, H., Yang, P.D., and Majumdar, A., Ion transport in nanofluidic channels, *Nano Letters*, 2004, vol. 4, no. 1, p. 137.
- Dukhin, S.S. and Mishchuk, N.A., Unlimited increase in the current through ionite granules, *Kolloidn. Zh.*, 1988, vol. 49, no. 6, p. 1047.
- Dukhin, S.S., Electrokinetic phenomena of second kind and their applications, *Adv. Colloid Interface Sci.*, 1991, vol. 35, p. 173.
- Mishchuk, N.A., Concentration polarization of interface and non-linear electrokinetic phenomena, *Adv. Colloid Interface Sci.*, 2010, vol. 160, p. 16.
- Rubinstein, I. and Zaltzman, B., Equilibrium electroconvective instability, *Phys. Rev. Lett.*, 2015, vol. 114, 114502.
- Zaltzman, B. and Rubinstein, I., Electro-osmotic slip and electroconvective instability, *J. Fluid Mech.*, 2007, vol. 579, p. 173.
- Wessling, M., Morcillo, L.G., and Abdu, S., Nanometer-thick lateral polyelectrolyte micropatterns induce macroscopic electro-osmotic chaotic fluid instabilities, *Sci. Rep.*, 2014, vol. 4, p. 4294.
- Dukhin, S.S. and Mishchuk, N.A., Intensification of electrodialysis based on electroosmosis of the second kind, *J. Membr. Sci.*, 1993, vol. 79, p. 199.
- Nikonenko, V.V., Vasil'eva, V.I., Akberova, E.M., Uzdénova, A.M., Urtenov, M.K., Kovalenko, A.V., Pismenskaya, N.P., Mareev, S.A., and Pourcelly, G., Competition between diffusion and electroconvection at an ion-selective surface in intensive current regimes, *Adv. Colloid Interface Sci.*, 2016, vol. 235, p. 233.
- Rubinstein, I., Staude, E., and Kedem, O., Role of the membrane surface in concentration polarization at ion-exchange membranes, *Desalination*, 1988, vol. 69, p. 101.
- Rubinstein, I. and Shtilman, L., Voltage against current curves of cation-exchange membranes, *J. Chem. Soc., Faraday Trans.*, 1979, vol. 75, p. 231.
- Rubinstein, I., Zaltzman, B., and Pundik, T., Ion-exchange funneling in thin-film coating modification of heterogeneous electrodialysis membranes, *Phys. Rev. E*, 2002, vol. 65, 041507.
- Belova, E.I., Lopatkova, G.Yu., Pismenskaya, N.D., Nikonenko, V.V., Larchet, C., and Pourcelly, G., The effect of anion-exchange membrane surface properties on mechanisms of overlimiting mass transfer, *J. Phys. Chem. B*, 2006, vol. 110, p. 13458.
- Dydek, E.V., Zaltzman, B., Rubinstein, I., Deng, D.S., Mani, A., and Bazant, M.Z., Overlimiting current in a microchannel, *Phys. Rev. Lett.*, 2011, vol. 107, 118301.
- Green, Y. and Yossifon, G., Dynamical trapping of colloids at the stagnation points of electro-osmotic vortices of the second kind, *Phys. Rev. E*, 2013, vol. 87, no. 3, 033005.
- Demekhin, E.A., Shelistov, V.S., and Polyanskikh, S.V., Linear and nonlinear evolution and diffusion layer selection in V.S. electrokinetic instability, *Phys. Rev. E*, 2011, vol. 84, 036318.
- Shelistov, V., Nikitin, N., Ganchenko, G., and Demekhin, E., Numerical modeling of electrokinetic instability in semipermeable membranes, *Dokl. Phys.*, 2011, vol. 56, p. 538.
- Pham, S.V., Li, Z., Lim, K.M., White, J.K., and Han, J., Direct numerical simulation of electroconvective instability and hysteretic current-voltage response of perm-

- selective membrane, *Phys. Rev. E*, 2012, vol. 86, 046310.
26. Druzgalski, C.L., Andersen, M.B., and Mani, A., Direct numerical simulation of electroconvective instability and hydrodynamic chaos near an ion-selective surface, *Phys. Fluids*, 2013, vol. 25, 110804.
 27. Dukhin, S.S., Electrokinetic phenomena of the second kind and their applications, *Adv. Colloid Interface Sci.*, 1991, vol. 35, p. 173.
 28. Mishchuk, N.A. and Takhistov, P.V., Electroosmosis of the second kind and current through curved interface, *Colloids Surf. A*, 1995, vol. 95, p. 119.
 29. Mishchuk, N.A., Electro-osmosis of the second kind near the heterogeneous ion-exchange membrane, *Colloids Surf. A*, 1998, vol. 140, p. 75.
 30. Chang, H.-C., Demekhin, E.A., and Shelistov, V.S., Competition between Dukhin's and Rubinstein's electrokinetic modes, *Phys. Rev. E*, 2012, vol. 86, 046319.
 31. Uzdenova, A.M., Kovalenko, A.V., and Urtenov, M.Kh., *Matematicheskie modeli elektrokonevtsii v elektromembrannykh sistemakh* (Mathematical Models of Electroconvection in Electrochemical Systems), Karachayevsk: KChGU, 2011.
 32. Nikonenko, V.V., Kovalenko, A.V., Urtenov, M.K., Pismenskaya, N.D., Han, J., Sistat, P., and Pourcelly, G., Desalination at overlimiting currents: state-of-the-art and perspectives, *Desalination*, 2014, vol. 342, p. 85.
 33. Uzdenova, A.M., Kovalenko, A.V., Urtenov, M.K., and Nikonenko, V.V., Effect of electroconvection during pulsed electric field electrodialysis. Numerical experiments, *Electrochem. Commun.*, 2015, vol. 51, p. 1.
 34. Uzdenova, A.M., Kovalenko, A.V., and Urtenov, M.Kh., *Matematicheskoe modelirovanie membrannykh protsessov s ispol'zovaniem Comsol Multiphysics 4.3* (Mathematical Simulation of Membrane Processes with the Use of Comsol Multiphysics 4.3), Krasnodar: KGU, 2013.
 35. Newman, J.S., *Electrochemical Systems*, New York: Prentice Hall, 1991, 2nd Ed.
 36. Gnusin, N.P., Zabolotskii, V.I., Nikonenko, V.V., and Urtenov, M.Kh., Convective-diffusion model of the process of electrodialysis desalination. Overlimiting current and diffusion layer, *Elektrokhimiya*, 1986, vol. 22, p. 298.
 37. Karatay, E., Druzgalski, C.L., and Mani, A., Simulation of chaotic electrokinetic transport: performance of commercial software versus custom-built direct numerical simulation codes, *J. Colloid Interface Sci.*, 2015, vol. 446, p. 67.
 38. Urtenov, M.A.K., Kirillova, E.V., Seidova, N.M., and Nikonenko, V.V., Decoupling of the Nernst-Planck and Poisson equations. Application to a membrane system at overlimiting currents, *J. Phys. Chem. B*, 2007, vol. 111, p. 14208.
 39. Belashova, E.D., Melnik, N.A., Pismenskaya, N.D., Shevtsova, K.A., Nebavsky, A.V., Lebedev, K.A., and Nikonenko, V.V., Overlimiting mass transfer through cation-exchange membranes modified by Nafion film and carbon nanotubes, *Electrochim. Acta*, 2012, vol. 59, p. 412.
 40. Rubinstein, I., Zaltzman, B., Futerman, A., Gitis, V., and Nikonenko, V., Reexamination of electrodiffusion time scales, *Phys. Rev. E*, 2009, vol. 79, no. 2, 021506.

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